# Predictive Analytics Using Historical Data

## Thiranex Data Analytics Internship - Task 3

## Objective

The objective of this project is to build a predictive analytics model using historical sales data and forecast future sales trends.

## Dataset

E-Commerce Sales and Profit Analysis Dataset containing Order Date, Product Name, Category, Region, Quantity, Sales, and Profit.

## Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Scikit-learn, and Jupyter Notebook.

## Methodology

1. Loaded and inspected the dataset.
2. Converted Order Date into datetime format.
3. Aggregated sales on a monthly basis.
4. Created a numerical time index for model training.
5. Trained a Linear Regression model.
6. Evaluated model performance using MAE and $R^2$ Score.
7. Forecasted future sales for the next six months.
8. Visualized historical and predicted sales trends.

## Results

Mean Absolute Error (MAE): 35,867.67
$R^2$ Score: 0.08

## Interpretation

The model demonstrated the process of predictive analytics and sales forecasting. The $R^2$ score indicates that sales variability is influenced by factors not captured by the simple linear regression model. Future predictions show a gradual upward sales trend.

## Conclusion

A predictive analytics solution was successfully implemented using historical sales data. The project provided practical experience in data preprocessing, machine learning, forecasting, and model evaluation. More advanced forecasting techniques such as ARIMA, Prophet, or ensemble models could improve prediction accuracy in future work.